Leng et al, Intracranial arterial stenosis in Caucasian versus Chinese patients with TIA and minor stroke: two contemporaneous cohorts and a systematic review

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eTable 1. Search strategy for PubMed, conducted on 14 October 2019

Search #	Search queries	Number of items found
#1	Search (intracranial arter* steno* OR intracranial steno* OR intracranial large artery steno* OR intracranial occlusi* OR intracranial arter* occlusi* OR intracranial large artery occlusi* OR intracranial atherosclero* OR intracranial large artery atherosclero*) [All Fields]	2,388
#2	Search (prevalence OR incidence OR frequency OR epidemiology OR burden) [All Fields]	3,831,047
#3	Search (ischemic stroke OR ischaemic stroke OR minor stroke OR minor ischemic stroke OR minor ischaemic stroke OR transient ischemic attack OR transient ischaemic attack OR TIA) [All Fields]	113,316
#4	Search #1 AND #2 AND #3	373
#5	Search #4 AND ("1990/01/01" [Date - Publication]: "2019/10/14" [Date - Publication]) AND English [Language]	324

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eTable 2. Search strategy for OVID, conducted on 14 October 2019 [a]

Search #	Search queries	Number of items found
#1	(intracranial arter* steno* or intracranial steno* or intracranial large artery steno* or intracranial occlusi* or intracranial arter* occlusi* or intracranial large artery occlusi* or intracranial atherosclero* or intracranial large artery atherosclero*).mp. [mp=ti, ab, hw, tn, ot, dm, mf, dv, kw, fx, dq, nm, kf, ox, px, rx, an, ui, sy]	8,452
#2	(prevalence or incidence or frequency or epidemiology or burden).mp. [mp=ti, ab, hw, tn, ot, dm, mf, dv, kw, fx, dq, nm, kf, ox, px, rx, an, ui, sy]	10,186,862
#3	(ischemic stroke or ischaemic stroke or minor stroke or minor ischemic stroke or minor ischaemic stroke or transient ischemic attack or transient ischaemic attack or TIA).mp. [mp=ti, ab, hw, tn, ot, dm, mf, dv, kw, fx, dq, nm, kf, ox, px, rx, an, ui, sy]	231,419
#4	#1 AND #2 AND #3	1293
#5	limit #4 to yr="1990 -Current"	1283
#6	limit #5 to (English) AND (full text) AND (human)	563

[a] The following databases were searched via OVID: Embase 1910 to Present; Ovid MEDLINE(R) 1946 to October 14, 2019; and Ovid MEDLINE(R) and Epub Ahead of Print, In-Process & Other Non-Indexed Citations 1946 to October 14, 2019.

eTable 3: Details of imaging received by study patients

Imaging modality	OXVASC	CUHK-SR
	n (%) N= 1579	n (%) N= 1099
Intracranial vascular imaging		
MR angiography	1035 (65.5)	673 (61.2)
CT angiography	253 (16.0)	99 (9.0)
MR and CT angiography	0	81 (7.4)
Transcranial Doppler only	81 (5.2)	262 (23.8)
Digital subtraction angiography only	0	1 (0.1)
Extracranial vascular imaging only		
Carotid Doppler ultrasound	154 (9.8)	38 (3.5)
No vascular imaging	57 (3.6)	107 (9.7)
Known atrial fibrillation	25	62
Previously investigated or imaged elsewhere	8	0
Other clinical or logistical issue[a]	24	45

Abbreviations: OXVASC, Oxford Vascular Study; CUHK-SR, the Chinese University of Hong Kong Stroke Registry.

[a] Such as older than 90 years, gastrointestinal bleeding, trauma, dementia.

eTable 4: Baseline characteristics of study population, stratified by intracranial vascular imaging.

Characteristics	OXVASC			CUHK-SR		
	Intracranial MRA/CTA (n= 1,287)	***No intracranial MRA/CTA (n= 292)***	**p-value**	***Intracranial MRA/CTA (n= 691)***	***No intracranial MRA/CTA (n= 408)***	**p-value**
Age, years	69 (13.9)	75 (12.8)	<0.0001	66 (12.3)	72 (11.9)	<0.001
Age ≥70 years	715 (55.6)	209 (71.6)	<0.0001	279 (40.4)	243 (59.6)	<0.001
Male sex	667 (51.8)	143 (49.0)	0.38	441 (63.8)	213 (52.2)	<0.001
Ever-smoker	688 (53.5)	162 (56.3)	0.40	270 (39.1)	154 (37.7)	0.662
History of hypertension	707 (54.9)	194 (67.4)	<0.0001	453 (65.6)	286 (70.1)	0.121
History of diabetes	169 (13.1)	62 (21.6)	<0.0001	181 (26.2)	116 (28.4)	0.420
History of dyslipidemia	436 (33.9)	109 (38.1)	0.17	354 (51.2)	169 (41.4)	<0.002
Atrial fibrillation [a]	191 (14.9)	70 (24.3)	<0.0001	78 (11.3)	95 (23.3)	<0.001
History of ischemic stroke/TIA	192 (14.9)	88 (30.4)	<0.0001	144 (20.8)	90 (22.1)	0.633
History of ischemic heart disease	166 (12.9)	65 (22.5)	<0.0001	54 (7.8)	52 (12.7)	0.007
Event type						0.015
TIA	842 (65.4)	158 (54.1)	<0.0001	208 (30.1)	152 (37.3)	
Minor stroke	445 (34.6)	134 (45.9)		483 (69.9)	256 (62.7)	

Abbreviations: OXVASC, Oxford Vascular Study; CUHK-SR, the Chinese University of Hong Kong Stroke Registry; MRA, magnetic resonance angiography; CTA, computed tomography angiography; TIA, transient ischemic attack.

[a] History of atrial fibrillation and newly diagnosed atrial fibrillation after the index stroke or TIA.

eTable 5. Risk factors of ICAS in younger and older patients in the 2 cohorts

Risk factors	OXVASC (n=1287)				CUHK-SR (N=691)			
	Patients <70 years old (n=572)		Patients ≥70 years old (n=715)		Patients <70 years old (n=412)		Patients ≥70 years old (n=279)	
	Multivariable adjusted OR (95% CI)	p-value	Multivariable adjusted OR (95% CI)	p-value	Multivariable adjusted OR (95% CI)	p-value	Multivariable adjusted OR (95% CI)	p-value
Age (every 10-year increment)	0.97 (0.72-1.30)	0.818	1.79 (1.37-2.35)	<0.001	1.00 (0.78-1.28)	0.973	1.86 (1.10-3.15)	0.020
Male sex	1.72 (0.94-3.16)	0.079	1.28 (0.90-1.82)	0.173	1.01 (0.65-1.55)	0.980	1.05 (0.63-1.74)	0.858
Ever-smoker	1.01 (0.57-1.79)	0.965	0.98 (0.69-1.39)	0.909	0.94 (0.62-1.43)	0.774	1.19 (0.72-1.98)	0.494
History of hypertension	1.68 (0.92-3.06)	0.090	1.48 (1.00-2.18)	0.048	1.40 (0.90-2.20)	0.138	1.68 (0.93-3.02)	0.086
History of diabetes	1.67 (0.77-3.61)	0.191	1.40 (0.88-2.23)	0.155	1.63 (0.99-2.66)	0.054	1.86 (1.09-3.18)	0.023
History of dyslipidemia	1.05 (0.55-2.02)	0.873	1.24 (0.87-1.77)	0.230	1.23 (0.80-1.87)	0.342	1.57 (0.96-2.59)	0.074

Abbreviations: ICAS, Intracranial atherosclerotic stenosis; OXVASC, Oxford Vascular Study; CUHK-SR, the Chinese University of Hong Kong Stroke Registry; OR, odds ratio; CI, confidence interval.

eTable 6. Prevalence of ICAS in patients with ischemic stroke or TIA in Europe

Study	Country/ Region	Enrolment period	Sample size	Mean age, years	Male, %	IS/TIA	Cerebrovascular imaging to define ICAS [a]	Cerebral vascular workups of the patients				Prevalence of ICAS, %		Modified NOS [b]
								TCD, % or n (%)	MRA, % or n (%)	CTA, % or n (%)	DSA, % or n (%)	Any ICAS	Symptomatic ICAS	
Weimar 2006 [1]	Germany	2000-2002	4,157	67	57	IS/TIA	TCD/MRA/ CTA/DSA	97	24	3	4	—	15	4
Holzer 2009 [2]	Germany	2000-2004	176	63	38	TIA	TCD	176 (100)	NA	NA	NA	12	—	2
Meseguer 2010 [3]	France	2003-2007	1,823	62	50	TIA	TCD/MRA/ CTA/HR-MRI	1,823 (100)	1,248 (69)	82 (5)	NA	9	4	3
Homburg 2011 [4]	Netherlands	—	786	62	56	IS/TIA	CTA	786 (100)	NA	NA	NA	10	3	3
Ovesen 2013 [5]	Denmark	2009-2011	652	67	56	IS/TIA	CTA	652 (100)	NA	NA	NA	7	0.5	3
von Sarnowski 2013 [6]	Pan-European countries	2007-2010	1,612	46	82	Young IS/TIA (age 18-55)	TCD	1,612 (100)	NA	NA	NA	20% in IS; 7% in TIA	14% in IS; 4% in TIA	4
Ssi-Yan-Kai 2013 [7]	France	2009-2011	129	64	57	Minor IS/ TIA	MRA	NA	129 (100)	NA	NA	—	12	2
Wolff 2014 [8]	France	2005-2010	159	37	51	Young IS (age 18-45)	TCD/MRA/ CTA/DSA	157 (99)	144 (91)	47 (30)	68 (44)	—	31	2
Logallo 2014 [9]	Norway	2010-2011	607	72	55	IS/TIA	TCCS/MRA/ CTA	TCCS: 607 (100)	533 (88)	227 (37)	NA	11	7	3

Tsivgoulis 2014 [10]	Greece	2009-2011	467	58	60	IS/TIA	TCD/MRA/ CTA/DSA	467 (100)	Only in those suspected of ICAS by TCD; numbers not reported			11	9	4
Baracchini 2016 [11]	Italy	2011-2013	1,134	71	59	IS	TCD/TCCS/ MRA/CTA	TCD/ TCCS: 1,134 (100)	844 (74)	290 (26)	113 (10)	9% (uncertain if it was any ICAS or symptomatic ICAS)		4
Sorgun 2017 [12]	Turkey	2012-2014	619	69	54	IS	MRA/CTA/ DSA	—	—	—	—	12	11	3
Hoshino 2018 [13]	France	2005-2008	403	62	74	IS	TCD/MRA/ CTA	399 (99)	289 (72)	53 (13)	NA	36	18	3

Abbreviations: ICAS, intracranial atherosclerotic stenosis or intracranial arterial stenosis of any etiology as defined in different studies; TIA, transient ischemic attack; IS, ischemic stroke; TCD, transcranial Doppler, MRA, magnetic resonance angiography; CTA, computed tomographic angiography; DSA, digital subtraction angiography; NOS, Newcastle-Ottawa Scale; HR-MRI, high-resolution magnetic resonance imaging; TCCS, transcranial color-coded Doppler sonography.

[a] ICAS was defined by velocity criteria in TCD or >/≥50% stenosis or occlusion in MRA/CTA/DSA, if not specified otherwise.

[b] With a total score of 0-5; scores of ≥3 or <3 respectively indicating low and high risk of bias.

"—" indicates unclear or not reported; "NA" indicates not applicable.

eTable 7. Prevalence of ICAS in patients with ischemic stroke or TIA in Asia

Study	Country/ Region	Enrolment period	Sample size	Mean age, years	Male, %	IS/TIA	Cerebrovascular imaging to define ICAS [a]	Cerebral vascular workups of the patients				Prevalence of ICAS, %		Modified NOS [c]
								TCD, n (%)	MRA, n (%)	CTA, n (%)	DSA, n (%)	Any ICAS	Symptomatic ICAS	
Lee 2002 [14]	Taiwan	1997-2001	170	—	71	Young IS/TIA (age 18-45)	DSA	NA	NA	NA	170 (100)	27	—	2
Shin 2005 [15]	South Korea	2000-2003	901	62	59	IS	MRA/DSA	NA	—	NA	—	28 [b]	22	3
Kim 2010 [16]	South Korea	2006-2009	1,012	64	63	IS/TIA	MRA	NA	1,012 (100)	NA	NA	27	—	3
Niu 2014 [17]	China	2007-2012	197	39	65	Young IS (age 15-49)	TCD/MRA/ CTA/DSA	160 (81)	MRA/CTA/DSA: 159 (81)			51	35	2
Shi 2014 [18]	China	2010-2012	351	47	72	Young IS (age≤55)	MRA/CTA/DSA	NA	315 (90)	24 (7)	12 (3)	35	—	3
Roy 2014 [19]	India	2007-2013	610	49	70	IS	Possibly MRA/CTA	NA	—	—	NA	—	36	3
Kiyohara 2014 [20]	Japan	2007-2012	693	69	62	TIA	MRA/CTA/DSA	NA	672 (97)	158 (23)	57 (8)	22		4
Ojha 2015 [21]	China	2007-2012	123	45	80	Young IS (age ≤50)	MRA	NA	98 (80)	NA	NA	38	—	2
Lee 2017 [22]	South Korea	2011-2016	516	68	52	IS	MRA	NA	516 (100)	NA	NA	43	—	3

Kim 2017 [23]	South Korea	2006-2009	1,081	64	62	IS/TIA	MRA/CTA/DSA	Mostly MRA (numbers not reported)				37	17	3
Park 2017 [24]	South Korea	2013-2013	9,506	66	61	Minor IS/ high-risk TIA	TCD/MRA/CTA/ DSA	6,990 (74)	7,494 (79)	3,255 (34)	945 (10)	—	25	4
Zhang 2019 [25]	China	2017-2018	207	62	66	IS	CTA	NA	NA	207 (100)	NA	65	—	3

Abbreviations: ICAS, intracranial atherosclerotic stenosis or intracranial arterial stenosis of any etiology as defined in different studies; TIA, transient ischemic attack; IS, ischemic stroke; TCD, transcranial Doppler, MRA, magnetic resonance angiography; CTA, computed tomographic angiography; DSA, digital subtraction angiography; NOS, Newcastle-Ottawa Scale.

[a] ICAS was defined by velocity criteria in TCD or >/≥50% stenosis or occlusion in MRA/CTA/DSA, if not specified otherwise.

[b] Prevalence of any ICAS was not reported in this article[15]. This number (28%) was retrieved from another article[26] reporting ICAS prevalence in this cohort but with a smaller sample size (n=512).

[c] With a total score of 0-5; scores of ≥3 or <3 respectively indicating low and high risk of bias.

"—" indicates unclear or not reported; "NA" indicates not applicable.

eTable 8. Prevalence of ICAS in patients with ischemic stroke or TIA in North and South America

Study	Country/ Region	Enrolment period	Sample size	Mean age, years	Male, %	IS/TIA	Cerebrovascul -ar imaging to define ICAS [a]	Cerebral vascular workups of the patients				Prevalence of ICAS, %		Modified NOS [c]
								TCD, % or n (%)	MRA, % or n (%)	CTA, % or n (%)	DSA, % or n (%)	Any ICAS	Symptomatic ICAS	
Sacco 1995 [27]	USA (multi-ethnic)	1990-1993	438	70	46	IS	TCD/MRA/ DSA [b]	75	—	NA	—	—	1% in Caucasians; 6% in African Americans; 11% in Hispanics	4
Wityk 1996 [28]	USA (multi-ethnic)	1993-1995	274	67	50	IS/TIA	TCD/MRA/ DSA	14	49	NA	14	24% in Caucasians; 22% in African Americans	9	3
Koch 2005 [29]	USA (Haitian)	1998-2002	126	60	52	IS	MRA/DSA	NA	MRA/DSA in 81 (64%)			—	9	3
Rincon 2009 [30]	USA (multi-ethnic)	1993-1997	714	60	37	First IS	TCD/MRA/ DSA [b]	54	18	NA	3	—	7	4
Lange 2018 [31]	Brazil	2012-2015	359	64	51	First IS	TCD/MRA/ CTA/DSA	315 (88)	MRA/CTA/DSA in 184 (51%)			—	5	3

Abbreviations: ICAS, intracranial atherosclerotic stenosis or intracranial arterial stenosis of any etiology as defined in different studies; TIA, transient ischemic attack; IS, ischemic stroke; TCD, transcranial Doppler, MRA, magnetic resonance angiography; CTA, computed tomographic angiography; DSA, digital subtraction angiography; NOS, Newcastle-Ottawa Scale.

[a] ICAS was defined by velocity criteria in TCD or >/≥50% stenosis or occlusion in MRA/CTA/DSA, if not specified otherwise.

[b] ICAS was defined by velocity criteria in TCD or >60% stenosis or occlusion in MRA/DSA in this study.

[c] With a total score of 0-5; scores of ≥3 or <3 respectively indicating low and high risk of bias.

"—" indicates unclear or not reported; "NA" indicates not applicable.

eTable 9. Prevalence of ICAS in patients with ischemic stroke or TIA in other countries/regions or worldwide

Study	Country/ Region	Enrolment period	Sample size	Mean age, years	Male, %	IS/TIA	Cerebrovascular imaging to define ICAS [a]	Cerebral vascular workups of the patients				Prevalence of ICAS, %		Modified NOS [c]
								TCD, % or n (%)	MRA, % or n (%)	CTA, % or n (%)	DSA, n (%)	Any ICAS	Symptomatic ICAS	
TIAregistry.org project [32, 33]	World-wide	2009-2011	4,583	66	60	Minor IS/TIA	TCD/MRA/CTA	67	46	15	NA	14% overall; 20% in Japanese; 13% in non-Japanese [b]	—	4
Moustafa 2013 [34]	Egypt	2011	143	62	59	IS	MRA	NA	143 (100)	NA	NA	67	44	3

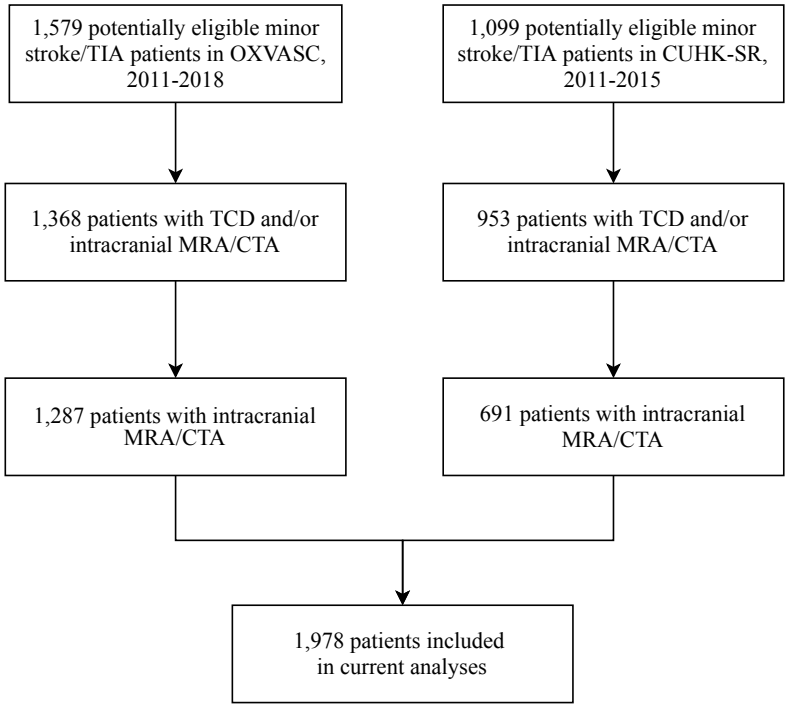
Abbreviations: ICAS, intracranial atherosclerotic stenosis or intracranial arterial stenosis of any etiology as defined in different studies; TIA, transient ischemic attack; IS, ischemic stroke; TCD, transcranial Doppler, MRA, magnetic resonance angiography; CTA, computed tomographic angiography; DSA, digital subtraction angiography; NOS, Newcastle-Ottawa Scale.

[a] ICAS was defined by velocity criteria in TCD or >/≥50% stenosis or occlusion in MRA/CTA/DSA, if not specified otherwise.

[b] 345 (7.5%) patients were Japanese; non-Japanese patients included European native (n = 3,317), Eastern Asian (n = 725), Middle or Proximal Eastern and Maghreb (n = 69), American Hispanic or Latino (n = 63), African American (n = 38), and other (n = 26).[33]

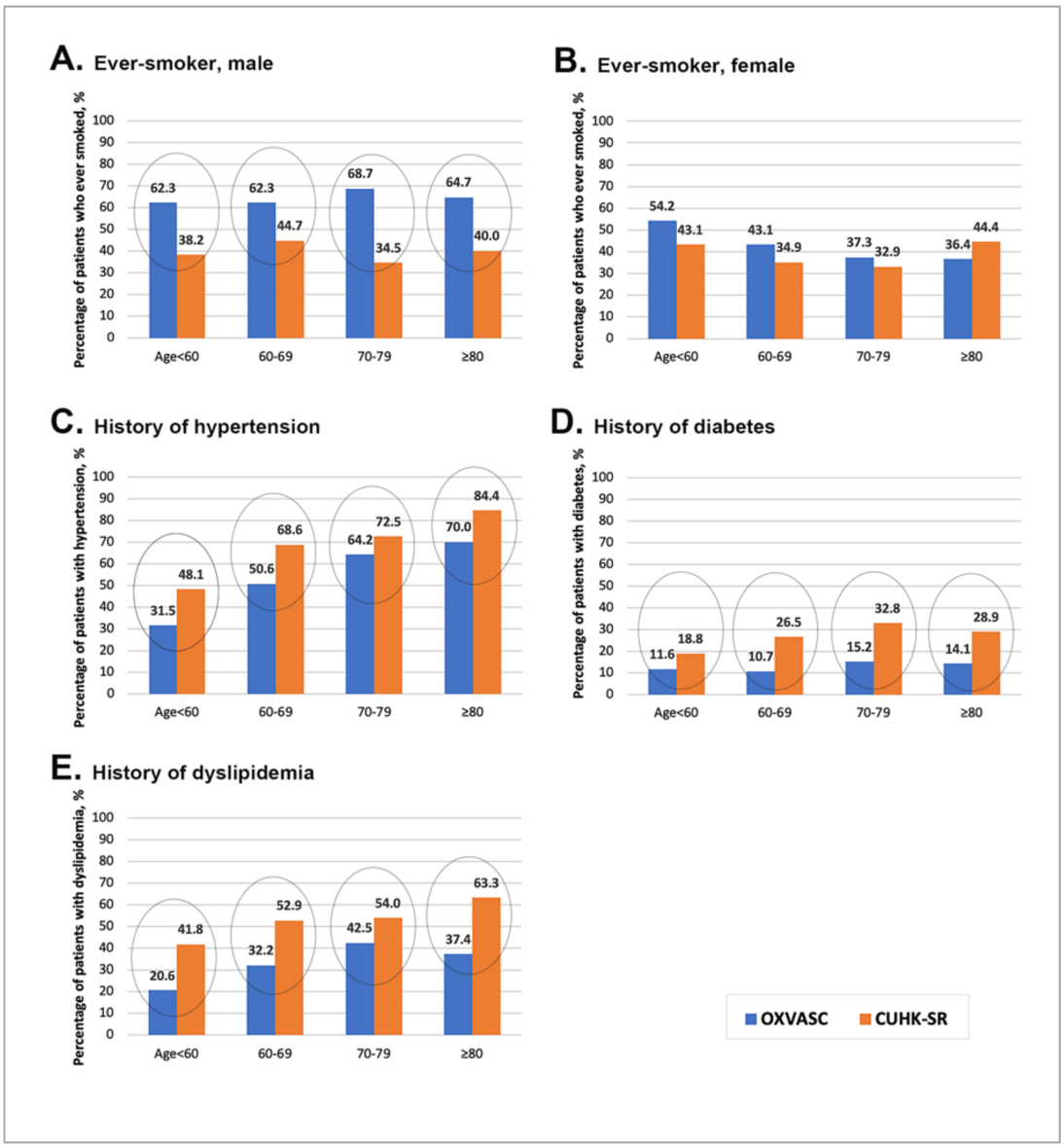
[c] With a total score of 0-5; scores of ≥3 or <3 respectively indicating low and high risk of bias.

"—" indicates unclear or not reported; "NA" indicates not applicable.



eFigure 1. Flow chart for patient screening in the current study.

Abbreviations: TIA, transient ischemic attack; OXVASC, Oxford Vascular Study; CUHK-SR, the Chinese University of Hong Kong Stroke Registry; TCD, transcranial Doppler; MRA, magnetic resonance angiography; CTA, computed tomography angiography.



eFigure 2. Age-specified prevalence of vascular risk factors in the 2 cohorts.

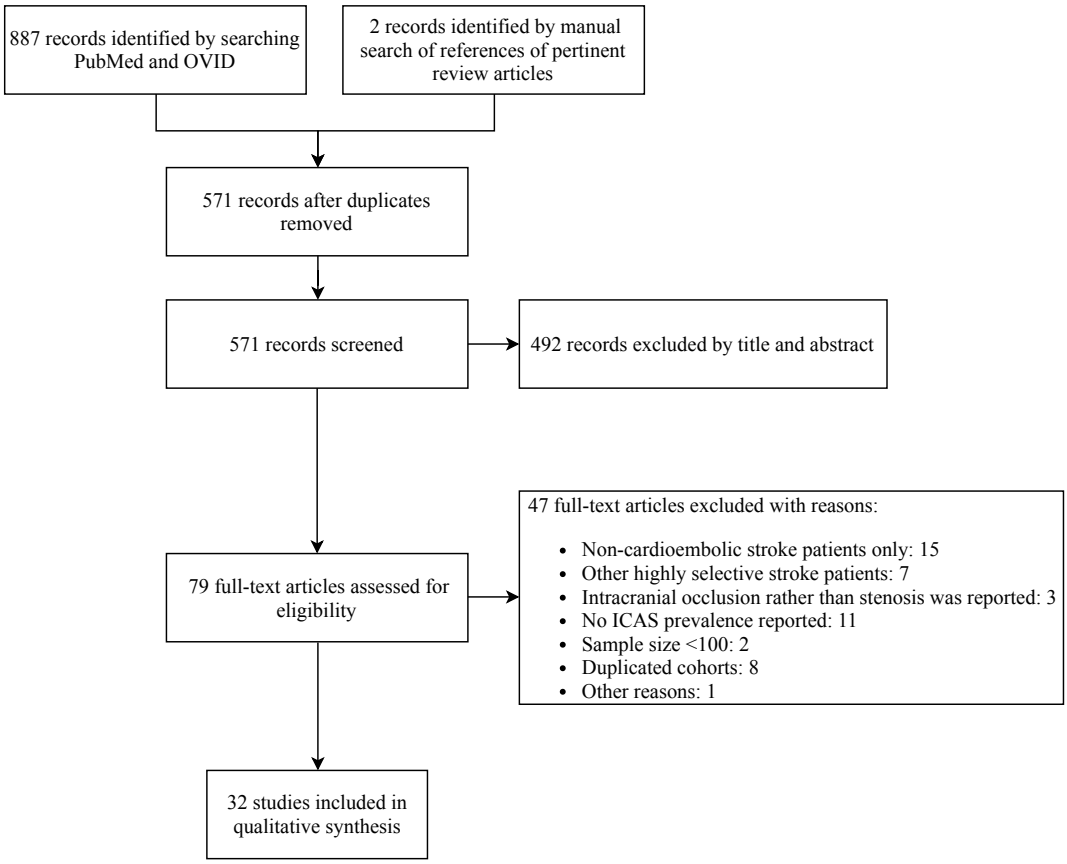
A. More male patients in OXVASC had ever smoked than those in CUHK-SR in each age group.

B. The percentages of female patients who had ever smoked in the two cohorts were not significantly different in any age group.

C/D/E. More patients in CUHK-SR had histories of hypertension, diabetes and dyslipidemia than those in OXVASC, in each age group.

Circles indicate comparisons with $p<0.05$ for chi-square tests.

Abbreviations: OXVASC, Oxford Vascular Study; CUHK-SR, the Chinese University of Hong Kong Stroke Registry.



eFigure 3. Flow chart for study screening and selection in the systematic review.

Abbreviations: ICAS, intracranial atherosclerotic stenosis.

Appendix. Modified Newcastle-Ottawa Scale (NOS) [35, 36] for quality assessment of the primary studies.

(1) Representativeness of the sample:

1 point: Population contained a mixture of specialties or at multiple sites.

0 points: Population contained a single specialty at a single site.

(2) Sample size:

1 point: Sample size was greater than 200 participants.

0 points: Sample size was less than 200 participants or a convenience sample.

(3) Non-respondents:

1 point: Comparability between respondent and non-respondent characteristics was established, and the response rate was satisfactory.

0 points: The response rate was unsatisfactory, the comparability between respondents and non-respondents was unsatisfactory, or there was no description of the response rate or the characteristics of the responders and the non-responders.

(4) Ascertainment of intracranial arterial stenosis (ICAS):

1 point: Independent assessment using objective imaging methods.

0 points: No description/non-standard imaging methods used.

(5) Quality of descriptive statistics reporting:

1 point: Reported descriptive statistics to describe the population (*e.g.*, age, sex) with proper measures of dispersion (*e.g.*, standard deviation, standard error, range).

0 points: Descriptive statistics were not reported, were incomplete, or did not include proper measures of dispersion.

Legend: This scale, ranging from 0 to 5, assesses quality in several domains: sample representativeness and size, comparability between respondents and non-respondents, ascertainment of ICAS, and statistical quality. Studies were judged to be of low risk of bias (≥3 points) or high risk of bias (<3 points).

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